Mini-Project Report Template: Characterization of a Non-Coding RNA Transcript

Course Assignment Deliverable Framework & Guided Workbook

Note to Students: Use this structural layout to record your execution steps, values, and biological justifications. Replace the italicized bracketed prompts with your actual computed data.

Part 1: Coding Potential Analysis

Tool Utilised: CPAT (Coding-Potential Assessment Tool) / CPC2 (Coding Potential Calculator 2)

Metric	Observed Value	Threshold / Reference (Human)
Coding Probability	[Insert Value, e.g., 0.032]	Thresh: 0.364 (mRNA >= 0.364)
Fickett Score	[Insert Value, e.g., 0.721]	Base composition asymmetry
Hexamer Score	[Insert Value, e.g., -0.145]	Codon usage bias indicator

Biological Justification & Verdict:

[Classify your transcript here as either a long non-coding RNA (lncRNA) or an unconventional mRNA. Justify this choice by demonstrating how your observed scores fall relative to the tool's empirical cut-off thresholds. Explain the mechanical role of hexamer bias and open reading frame (ORF) constraints.]

Part 2: Evolutionary Conservation Analysis

Browser Reference: UCSC Genome Browser / BLAT Alignments

- **PhastCons Score Evaluation:** [Examine base-by-base conservation probabilities. Document high vs. low probability clusters across vertebrate classes.]
- **PhyloP Score Evaluation:** [Document acceleration or deceleration rates. Interpret if specific loci show purifying evolutionary selection constraints.]
- **Evolutionary Scope Verdict:** [State if the transcript is broadly conserved across distant mammals/vertebrates, restricted to primates, or highly species-specific. Discuss the operational implications for functional vs. neutral evolution.]

Part 3: RNA Secondary Structure Prediction

Tool Utilised: RNAfold Web Server (ViennaRNA Package)

- **Minimum Free Energy (MFE):** [Insert value, e.g., -142.3 kcal/mol]
- **Thermodynamic Ensemble Partition Function:** [Insert ensemble free energy and frequency of the MFE structure]

Identified Stable Sub-Structures:

1. **Hairpin/Stem-Loop Region A:** Loci [Start-End]. Description of base-pairing stability and potential structural roles (e.g., safeguarding transcript stability against exonucleolytic breakdown or acting as a platform for protein recruitment).
2. **Hairpin/Stem-Loop Region B:** Loci [Start-End]. Potential biological significance (e.g., scaffolding complexes or hosting processing sites).

Interpretation & Integrative Conclusion

[Synthesise your structural, coding potential, and evolutionary observations. Highlight that while structural thermodynamic parameters suggest biochemical stability, actual physical interactions with RNA-binding proteins (RBPs) must be verified via empirical wet-lab protocols such as RIP-seq or CLIP-seq.]

This is for informational purposes only. For medical advice or diagnosis, consult a professional. AI responses may include mistakes.